

QUIZ 5 SOLUTIONS

Part A. ECBC-MAC uses two *independent* keys: K_{in} for the inner CBC-MAC and K_{out} for the final AES evaluation, i.e., $\text{ECBC-MAC}_{K_{in}\|K_{out}}(M) = \text{AES}_{K_{out}}(\text{CBC-MAC}_{K_{in}}(M))$. By contrast, $\mathcal{T}$ uses the *same* key K for both the inner CBC-MAC and the outer AES call.

Part B. We give an adversary that makes two **Tag** queries and achieves advantage 1.

Adversary $\mathcal{A}$:

Pick any $m \in \{0, 1\}^{128}$

$T_1 \leftarrow \text{Tag}(m)$

$T_2 \leftarrow \text{Tag}(T_1)$

Return $(m\|0^{128}\|0^{128}, T_2)$

Analysis. We verify that $\mathcal{A}$'s forgery is valid. Let $E = \text{AES}_K$ for ease of notation. For a one-block message x , $\text{CBC-MAC}_K(x) = E(x)$ (the IV is 0^{128}), so $\mathcal{T}_K(x) = E(E(x))$. Thus $T_1 = E^2(m)$ and $T_2 = E^2(T_1) = E^4(m)$. Now compute $\mathcal{T}_K(m\|0^{128}\|0^{128})$:

$$\begin{aligned} C_1 &= E(m), \\ C_2 &= E(C_1 \oplus 0^{128}) = E^2(m) = T_1, \\ C_3 &= E(C_2 \oplus 0^{128}) = E(T_1), \\ \mathcal{T}_K(m\|0^{128}\|0^{128}) &= E(E(T_1)) = E^2(T_1) = T_2. \end{aligned}$$

As forgery message $m\|0^{128}\|0^{128}$ has *three* blocks, while the two queries m and T_1 are *single blocks*, the forgery is “fresh” (i.e., the forgery message was never queried to **Tag**). Hence,

$$\text{Adv}_{\mathcal{T}}^{\text{uf-cma}}(\mathcal{A}) = 1.$$

Moreover, $\mathcal{A}$ is efficient since it makes only two queries and performs minor additional computation. **Why this fails for ECBC-MAC.** It is instructive to see how the attack breaks down for ECBC-MAC. With independent keys, the step $T_2 = \mathcal{T}_{K_{in}\|K_{out}}(T_1) = E_{K_{out}}(E_{K_{in}}(T_1))$, whereas $\mathcal{T}_{K_{in}\|K_{out}}(m\|0^{128}\|0^{128}) = E_{K_{out}}(E_{K_{in}}^3(m))$. These are equal only if $E_{K_{in}}(T_1) = E_{K_{in}}^3(m)$, i.e., $T_1 = E_{K_{in}}^2(m)$. But $T_1 = E_{K_{out}}(E_{K_{in}}(m)) \neq E_{K_{in}}^2(m)$ in general, so the attack breaks down.